



Profiling the biomechanics and risk of injury to the upper body during an over arm American football throw.

BEng PROJECT FINAL DISSERTATION 2023-24

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Summary

There is a large lack of research into the throwing technique in American Football even though it is a very fast-growing sport with millions of people learning to play every year. The throwing technique is often compared to sports like pitchers in baseball, however there are differences in the throwing technique between the two sports. For example, American football players release the ball a lot sooner than a baseball pitcher and have a more upright position, so it is important to research specifically into American football to understand the specific technique they use. This project aimed to analyse the movement of the shoulder and elbow while a quarterback completes a basic overarm throw of an American football at three different distances, in total six participants movements were analysed as a part of the study. Data was collected on participants planar movements, like elbow extension, for comparison to movements when completing a throw. It was deduced whether the experience of a player reduces or increases the risk of injury in the elbow or shoulder due to exceeding the range of motion in these joints. When playing American football quarterbacks have to throw a range of distances so the study has looked at three differences that will be comfortable and challenge participants. This allows the study to conclude whether the distance thrown means participants change their technique. The Xsens motion capture suit was used in the study to look at the biomechanics of different participants movement as it is currently one of the best motion capture technologies on the market. The project concluded that beginners and advanced

players are most likely to be at risk of injury through an overarm American football throw. Beginners were found to have continuous overextension of the elbow joint for every distance of throw, however, advance players were likely to get injured when completing longer throws due to them over rotating their shoulder joint in the aim to get more movement in the ball so it can travel the further distance. A suggestion for completing a basic throwing drill before training and games will help reinforce the correct technique for beginners to reduce their risk of injury.

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1. INTRODUCTION.

American football is a sport widely watched, with 115 million people watching the Superbowl in 2023 just in the US [1], however there is little research into American football around the world as it is not seen as a global sport unlike sports like football. Since the sport is not seen as a universal sport there is a void in research into quarterbacks' technique specifically and why they can experience pain in their upper body joints after playing. This project will explore the motion taking place at the joints of the shoulder and elbow during a throw and whether this has an impact on the pain some quarterbacks experience.

In the NFL (National America Football League) shoulder injuries are the second most common injury behind head injuries [2], and 14% of shoulder injuries come from the throwing motion rather than contact on the quarterback from defensive players [3]. Due to the lack of research into a quarterback's technique research on other sports which involve overarm throws must be looked at so that an optimal technique to avoid a higher risk of injury can be established. This is not always useful as when comparing a baseball pitcher and quarterback, quarterbacks stand in a more upright position and take shorter strides than a baseball pitcher [4]. Quarterbacks also must release the ball sooner to avoid defenders, leading to a lower mechanical load across the arm meaning they assume a position with a larger horizontal adduction. Although there is limited research into American football over arm throws it does still have the same basics of an overarm throw, these fundamental stages are, wind-up, stride, early cocking, late cocking, acceleration, deceleration, and the follow through, a basic model of this technique is shown in Figure 1 [5]. For all sports with overarm throws involved shoulder injuries are the one of most common injuries experienced by participants [6]. Elbow injuries also do occur to quarterbacks and even though they are less common than shoulder injuries it is also important to look at the elbow in the research [7].

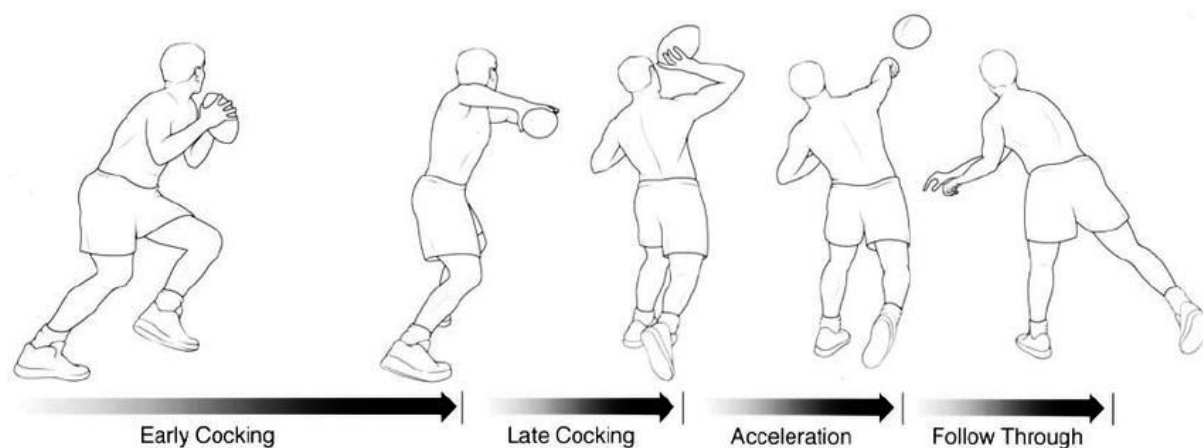


Figure 1 - Stages of a throw [8]

This project looks at female participants in particular, in American women's American football is only watched by a few hundred thousand people, however especially in European countries women's flag American football is a fast-growing sport and the sport has just been announced to be a sport in the 2028 Los Angeles Olympics. In addition, the NFL launched a program to

bring flag American football into British schools and now five hundred schools and thirty thousand players now participate in the sport, 53% of these players are female [9]. However, to make sure this project has as big of an impact on the sport as possible in helping players understand their pain and technique the study is only looking at the arm and shoulder as males and females both exhibit the same motion of an overarm throw in the arm and shoulder but present different movements in the hips and trunk [10].

After discussing with quarterbacks from the American Football Society at the University of Nottingham many of them said they experiences elbow pain or cracking/popping sounds at the joints especially after some trainings and games, which are symptoms of hyperextension of the joints. This study will find out if this is due to a difference in technique, or whether some players hyperextend their joints at the elbow and shoulder [11]. By taking the measurements of the participants maximum elbow extension and shoulder movement during planar movements and then in an overarm throw it can be seen if players overextend their arm, causing a greater risk to injury [12], and whether this has a correlation to a player's experience. The study looks at three groups of people with different experience levels: beginners to the sport (playing for less than six months), intermediate players (playing for six months to a year), and advanced players (playing for over a year). It is believed that beginners are more likely to get injured than advanced players as since they have not yet got the correct technique, in a study completed in 2018 it showed that novice runners reported more injuries per one thousand hours of running than advanced runners [13]. This project aims to show that like in baseball teaching and taking the time to understand the throwing motion [14] is vital in avoiding injury and prolonging a players career [15]. The study was expected to show results that the advanced players would have the best and most consistent techniques due to their experience and consistent coaching, it was also believed that beginners would be the most inconsistent across the three distances due to them not being as comfortable with the fundamentals of the technique.

1.1. Aim and Objectives

This project aims to analyse the biomechanical movement at the shoulder and elbow when a quarterback performs an overarm throw in American Football and analyse the risk of injuries at these joints.

Objectives:

- To record and analyse the biomechanical movement in the shoulder and elbow during an American Football throw at different distances.
- To look at participants planar movements, like elbow extension and then compare this to their elbow extension when completing a throw to look at whether players overextend their elbows during different distances of throws.
- To analyse the different movements and group the participants into three groups by the experience that they have (participants with experience less and six months,

participants with experience more than six months but less than a year and participants with experience of more than a year) and averaging the data of participants within each group.

- Conclude whether certain techniques cause a greater risk for injury to occur at the shoulder or elbow of a quarterback.

2. LITERATURE REVIEW

This study of literature will critically look at the body of knowledge regarding the technique of a quarterback during an overarm throw and injuries that can occur during this. A noticeable void in the existing research currently available is the exact technique of a quarterback, therefore the basic fundamentals of an overarm throw must be looked at, these fundamentals are consistent throughout most ball sports. A study which looks at a baseball pitcher, in the journal for Arthroscopy, Sports Medicine and Rehabilitation focuses on a baseball pitcher and gives a good insight into the fundamentals of an overarm throw breaking the technique down into a 6-step process of the wind-up, stride, cocking, acceleration, deceleration, and the follow through that is the same in American football [14]. An article in the Journal of Applied Biomechanics that compares high school and university baseball pitchers and quarterbacks found out the key differences in technique of the overarm throw in baseball and American football. It is suggested that during an overarm throw a quarterback is in a much more horizontal position in comparison to a baseball pitcher, and that quarterbacks make smaller steps during the motion [4]. This will lead to little differences in the shoulder joint and elbow joint angles between a baseball pitcher and American football quarterback, the difference in lengths of steps during the throw mean differences in the cocking stages in the 6 steps described in the study talked about in the Journal for Arthroscopy, Sports Medicine and Rehabilitation. Although, among the key ideas presented in both of these studies it does present that there are no major differences of fundamental techniques for an overarm throw and therefore we can use the ideas presented as the ideal technique for the participants in this study.

In the article “Kinematic and Kinetic Comparison Between Baseball Pitching and Football Passing” the ages of the participants are similar age to the participants in this study, so they should also have the same level of experience [4], however, all the participants are males, whereas this study is aiming to recruit males and females. This is the case for most of the other studies found while researching. This project involves looking at all genders movements, therefore before starting the experiment research had to be completed around the difference in techniques, it was found that the key difference between the techniques of men and women when completing an overarm throw was the rotation of the hips. In the article “Gender Differences in Motor Skills of the Overarm Throw” it was discussed that between the two genders they had the same movement in the humerus and forearm in the overhead throw of the handball but different movement in the trunk, leading this study to focus on the shoulder

and elbow so that the outcomes of the study will benefit players and coaches of all genders [10]. However, in that article just mentioned from the journal, Current Reviews in Musculoskeletal Medicine, the participants are ages 6-16 whereas this study plans to use players that are aged 18-22 so that all participants have a similar muscle and bone structure [6].

A study that was completed over 15 years from 2003 until 2018, which was published in 2020 looked into common injuries in youth American football quarterbacks, it was found that in a total of 374 quarterbacks there were a total of 423 injuries due to the sport. The top 5 locations that injuries that occurred were shoulder (22%), knee (15%) head/neck (14%), elbow (13%), and wrist/hand/lower arm (11%) [16]. Therefore, since this research showed that the elbow and shoulder were responsible for 35% of injuries sustained by the quarterbacks it highlights the importance of this research study in understanding how these injuries occur from the technique. However, similarly to other studies looked at while researching the participants in the study were under the age of 18, this means injuries are more likely to occur in youths due to children still learning to understand their physics limits [17].

Due to this study looking at whether certain techniques or abilities have a greater risk of injury there was lots of research looked at around injuries to quarterbacks specifically, there were three studies about injuries to quarterbacks that link to this study; common injuries in professional football quarterbacks [6], shoulder injuries to quarterbacks in the national football league [2] and acute elbow injuries in the national football league [7]. These three articles gave a good insight into where the most injuries occur for quarterbacks, and how they came to conclusions about the causes of the injuries. Similarly, to this study the internal and external rotation of joints and flexion/extension of the joints were analysed. However, these three articles all looked at professional athletes who have been playing at a very high level all of their lives, whereas many of the participants taking part in this study play the sport for fun and do not train or play to the same number of minutes as these professionals do.

Once data had been collected it was clear which participants over or under extend their elbow or shoulder joint, and if there is hyperextension of the joints. Hyperextension is injury when the joint is extended beyond its normal range of movement [11]. The normal range of movement at the elbow is 0° to 140° and at the shoulder the normal range of motion is flexion and abduction 180° with extension and adduction 50°, internal and external rotation at the shoulder with abducted arm 90° [18]. If the joint is extend past its normal range of motion it can cause the tissue around the joint to become damaged, causing pain, swelling and muscle spasms [12]. The shoulder is one of the most mobile joints in the body, but it is also one of the most unstable, when the joint is continuously over-rotated injury can occur. Hopefully this study can identify if players are at risk of injury and to what extent players hyperextend so that coaches and players can adapt techniques to prevent injury.

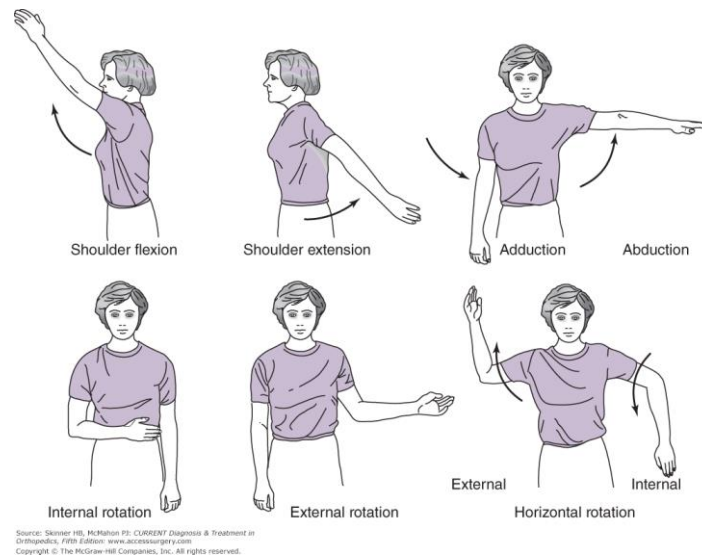


Figure 2 - Movements collected during the study [19]

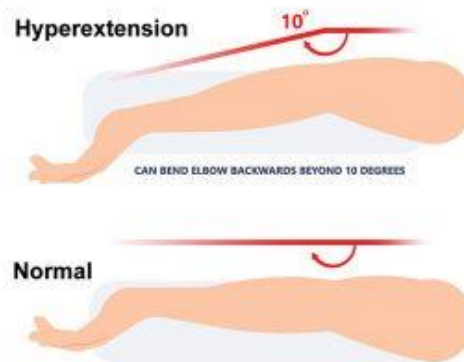


Figure 3 - Hyperextension at the elbow [20]

Due to recent developments in technologies in motion capture suits and the suits becoming more readily available, some studies which took place a few years ago would have preferred to have gotten conclusions from these motion capture suits, like what will be taking place in this study. In previous years video motion capture cameras were used to calculate joint angles and internal and external rotation as mentioned in the study Biomechanical Analysis of the Throwing Athlete and Its Impact on Return to Sport [5]. Recently there have been studies conducted to find out the reliability of the Xsens motion capture suit. A study was conducted in 2013 found that the MVN suit was most accurate in measuring flexion and extension of joints, and less accurate for the other two joint axis [21]. This enables this study to as accurate as possible as this study focuses on the flexion and extension of joints and whether some experience groups hyperextend at their joints leading to a larger risk of injury further into the future. However, since the study of the accuracy of the MVN suit took place eleven years ago, Movella have made more up to date equipment and listened to feedback to make the suit more reliable for information about movements that take place in the body [22]. So, the main source

and most up to date source of information on the motion capture suit be used to collect data with, is from the company that makes the suit, Movella. However, we have to be sceptical and cautious in believing in what is said by Movella, since they are the company that makes and sells the suits. As they are trying to sell their product to make profits for their business which could mean their information is bias on how good and how well the suit works.

3. METHODOLOGY

To analyse the biomechanical movement of a quarterback, throw the study recruited a total of six participants. The participants were required to complete two throws at three different distances and also complete planar movements of the arm, all of this was completed while wearing the Xsens motion capture suit by Movella. This allows the recording of the movements performed by participants. These recordings can be analysed using Movella software after data collection, the software will allow users to measure joint angles at the elbow and shoulder. Using this information there can be a comparison of the if there is any correlation between the chances of injury and experience of participants. It can also be seen which throws are more likely to cause injury, for example are participants overextending their arm more for longer throws as they require more power.

3.1. Ethical Considerations

Before testing could begin the study had to be ethically approved by the University of Nottingham's Research Integrity and Ethics Committee due to the use of human participants in the study. The ethical review was necessary to protect participants and the research team, it highlighted risks of the study and allowed a plan to be comprised to make sure participants knew what would happen to the data, knew what the study involved and how to leave the study if they wished to do so at any point. As part of this ethical review a risk assessment was conducted, this made sure that any risk could be flagged before data collection began allowing data collection to be planned to avoid the risks. For example, to reduce the risk of injury to participants, a full warm up was conducted before data collection began. All this information was presented to participants when they were invited to join the study. To ensure that the Data Protection Act 2018 was adhered to each participant was given a code in which all their data was stored under and are referred to throughout the study. The first letter refers to gender, F for female and M for Male, the number refers to their experience level, 1. experience less than six months, 2. experience more than six months but less than on year and 3. with experience more than a year, and the final letter indicated the participant they are within the group, A, B, C etc. for example the first female who has experience less than 6 months is known as F1A. In addition to this a risk assessment was completed before any data was collected.

3.2. Sampling

Participants were recruited from the University of Nottingham American Football Society, specifically the Flag American Football team. After speaking to coaches on the team, it was highlighted that all players on this team had basic level training on how to throw, but experience ranged from around three months to a year and a half of playing. This aligned with the levels of experience this study wanted to explore. Recruiting from people that had the basic knowledge to partake in the study meant that no time needed to be spent teaching recruits the movements we required them to complete. It would also reduce the risk of participants getting injured during the experiment as they knew the correct technique, this was important to consider to ensure the study could focus time on collecting data rather than participants have the basic fundamentals of a throw. Around thirty players that had completed the basic quarterback throwing session were contacted via email to participate. Ten of these players came back and showed interest in participating in the study. Each of these participants were placed into three experience groups, these groups were; participants with experience less and six months (three initial interested participants), participants with experience more than six months but less than a year (four initial interested participants), and finally participants with experience of more than a year (three initial interested participants). All of these participants willing to participate in the study turned out to be female, but this will not affect the overall conclusions of the study as males and females although having different movements in the hips and shoulders have the same movement in the arm during a throw. By the end of the study there was a total of six participants that took part, two participants with experience less than six months, three participants with experience more than six months but less than on year and one participant with experience more than a year. From each of these participants, data on their, age, height, shoes size and weight were collected to ensure the software had enough information for data to be accurate.

3.3. Data Collection

To collect data the Movella Xsens MVN Link was used. Xsens is a 3D motion tracking technology that allows biomechanical tracking of the body during sport activities. The MVN Link is an industry leading motion capture suit. The suit allows for high impact and highly dynamic movement, meaning that it is perfect for looking at the movement that will take place in the study. The material of the suit is Lycra, which has great elastic properties and shape retention making participants comfortable and unrestricted by the technology while completing the study. The 150m wireless range will allow for data to be collected no matter where on the pitch data was collected. The suit is a skin-tight suit that has 17 sensors on different parts of the body to make sure the movements are profiled to a high standard [23]; the sensors are small and lightweight so that they do not affect the movement of the individual wearing the suit [22]. For the study the sensors on the forearm, upper arm and shoulder were used. The sensors were easily put onto the suit before the participant arrived, and the participant just put the suit on before commencing the experiment, allowing for quick set-up. Each sensor can measure the acceleration and angular velocity during the movement.

Another option available to collect data was motion capture cameras. However, it was deemed that the Xsens suit was seen as more suitable for data collection in the study. If the cameras had been used, all data collection would have to take place in the laboratory. Even though the laboratory space is quite big, there was not enough space for participants to throw the ball. In the laboratory there is also lots of computer equipment that could be damaged if a participant had overthrown a ball. Using the suit meant that data collection could take place outside where there was lots of space, and this allowed the study of how throwing at different distances affected the technique of a participant.



Figure 4 - Suit used to collect information, photo was taken during practice data collection.

Before collecting data, all participants were required to do a warmup, and practice throws at each length. After this data on their weight and height was collected. The participant then had more practice throws with the suit on to make sure they were comfortable, and to check they didn't feel like the suit was restricting their movement due to it being the wrong size and the sensors were all in the correct places. To ensure consistency each participant was instructed to not move any sensors on the suit after the experiment had begun. If a sensor was moved this could cause different joint angles to what took place in the actual movements. The same ball was used with the same inflation for each session data was recorded to keep as many consistencies as possible, it was ensured that the weather was similar each time data was collected from a participant and the same location with markings was used with each participant. The participants first had their planar movements recorded, and then data of two throws were collected at the three distances respectively: ten yards, twenty yards and thirty yards. The majority of passes during games are within these three distances in women's flag games and men's games. The pitch that was being used for data collection had clear marks for each length used in the study allowing the participant to easily be and throw to the correct length required, which is why the distances were measured in yards rather than the traditional measurement of distance, meters.

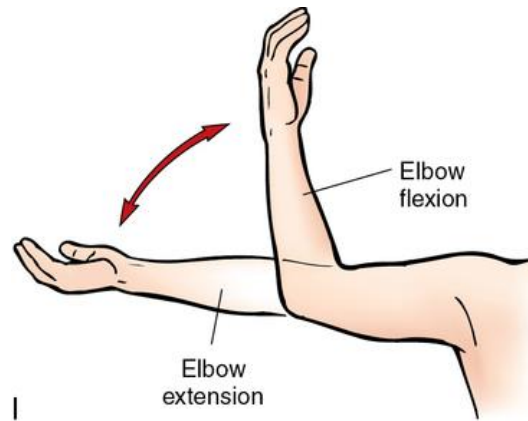


Figure 5 - Planar movement at the elbow performed by participants [24]

3.4. Data Analysis

Once all the data was collected the Xsens MVN software was used to analyse the data. Since the recorded data was in the x, y and z planes when analysing the data, information was gathered on the angular data; adduction/abduction, external/internal rotation and flexion/extension of the arm and shoulder. All this data was plotted on an easy-to-read graph. Using this graph provided by the software it could clearly be seen where the maximum and minimum angles were for each category. For each distance two sets of movements were recorded. This allows an average angle to be determined for each participant's movement. Gathering a mean value for each movement data for each participant gives an indication of the centre value located in the dataset. Using these mean angle values from each distance these will be compared to that presented in the planar movement, and a percentage of movement will be resolved using the equation:

$$\text{percentage of participants range of motion} = \frac{\text{angle during throwing motion}}{\text{angle at planar movement}} \times 100$$

Originally there was going to be a comparison between angles, however it was realised after the practice data was analysed that each participant would have a different range of movement at their joints. Finding the percentage provides a way to normalize that data across the different ranges of motion each participant has. Expressing the results as percentages shows the extent to which the participant is over/under extending/flexing their joints during the movement relative to themselves. This made comparison between groups of experience levels easy to read and make conclusions from. The data for this was taken from the player bringing their arm back (late cocking stage from Figure 1) until the follow through stage of the throw.

3.5. Changes and Reflection

At the end of data collection only five out of the original participants who agreed to join the study could make the times to collect data. This was due to reasons such as injury occurring outside of the study so they could not complete the movements, timetabling clashes and weather reducing the amount of data collection days. Upon reflection, more participants should have been invited into the study so that if one participant got injured before data collection took place there would still be other participants that could be asked to fill the injured participants' slot.

Due to software issues initially, each participant had ten throws recorded and an average would be taken from throws three through to seven, however data was lost after each participant second or third throw, so averages have been taken from the first two throws. This should not affect the overall outcome of the study, but if the mean could have been taken from ten throws and the median of each dataset calculated, outliers could have been identified and disregarded more easily than only have two movements in each dataset. In addition, if more participants data had been collected, mean and medians calculated, those participants who's mean and median did not correlate could have been removed from the study as it is preferred that throws are consistent throughout the dataset.

4. RESULTS AND DATA ANALYSIS

The results stage of the project was done using the Xsens MVN 2021.0.1 software produced by Movella. The program took the movements recorded of participants while wearing the suits and produced values for each joint. The focus of analysis was on the right elbow and shoulder joints of all participants, this was because everyone that took part on the study was right-handed, therefore, the main movement of the throw took place in this arm. The software produced a graph, as shown at the bottom of Figure 6, that shows the elbow extension and the shoulders, extension, flexion, abduction, internal rotation and external rotation. The graphs are measured in degrees, with zero degrees representing the body line when the shoulder and elbow is placed flat against the body. As the shoulder or elbow moves away from the body line created by the software the values increase.

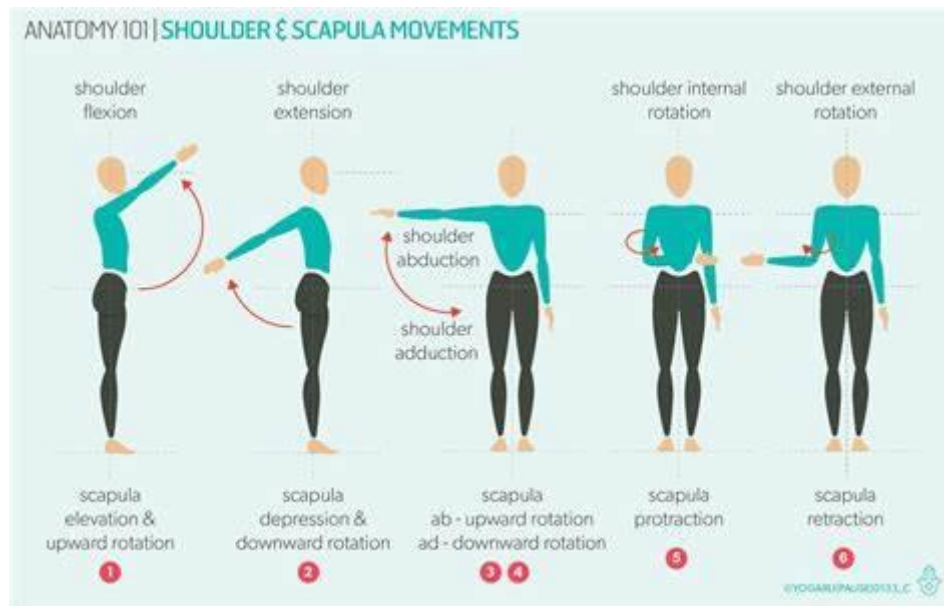


Figure 6 - Movements recorded at the shoulder [25]

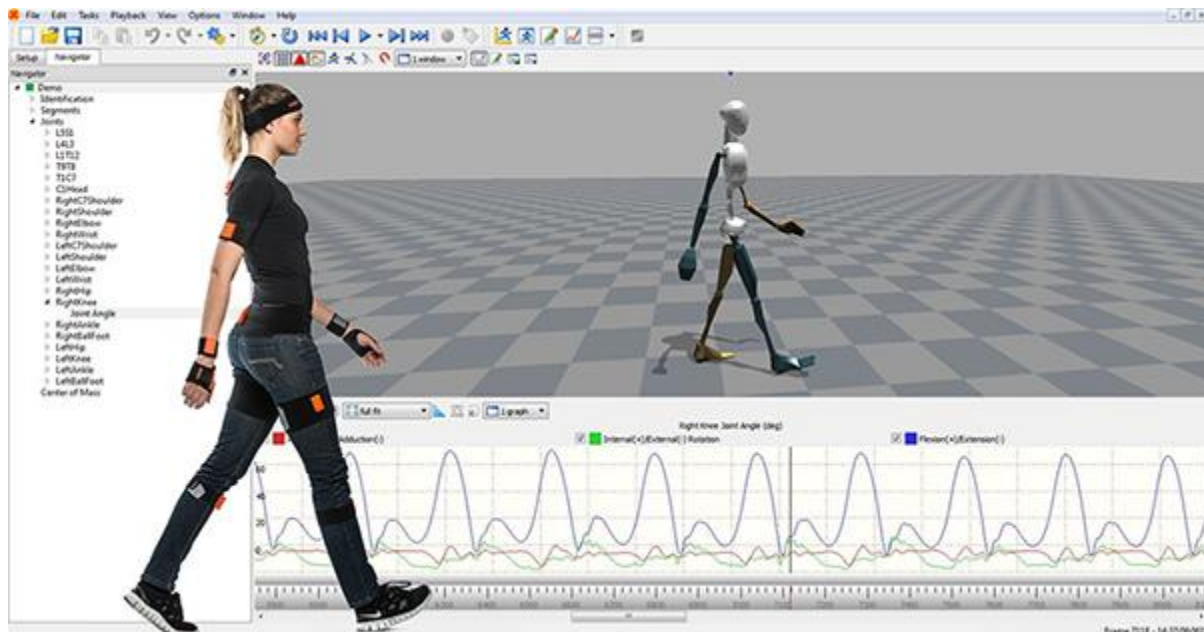


Figure 7 - Example of graph produced from walking movement [26]

When taking values, the maximum and minimum values for flexion/extension at the shoulder and elbow, abduction and internal/external rotation of the shoulder were located off the graph by highlighting the peaks and dips in the graphs as can be seen at the bottom of Figure 6. For each participant an average value between the two throws completed were calculated using Equation 1, then percentage of the participants average normal range of motion was calculated using Equation 2. Then finally the average range of motion of each group was calculated using Equation 3.

Equation 1:

$$\text{average angle for each participant} = \frac{\text{sum of angles}}{\text{number of throws completed}}$$

Equation 2:

$$\begin{aligned} &\text{percentage of participants range of motion} \\ &= \frac{\text{maximum angle during throwing motion}}{\text{maximum angle at planar movement}} \times 100 \end{aligned}$$

Equation 3:

$$\text{average percentage of range of motion} = \frac{\text{sum of average range of motion percentage}}{\text{number of participants in group}}$$

The primary research findings will be provided in this section. It starts with a chapter on the analysis of elbow joint. For the elbow joint, there was a focus on only the extension of the elbow as it a basic joint in comparison to the shoulder joint. Most injuries at the elbow occur due to the hyperextension of the joint. After analysing the elbow, the shoulder joint analysis was broken down into three subsections, flexion/extension, abduction and internal/external rotation. This is due to the complexity of the shoulder joints with its six degrees of freedom and the joint being the most instable in the human body. The results show some commonalities and differences to the hypothesis state in the introduction.

4.1. Elbow Extension

Table 5 (appendix) represents the angle that each participant over or under extends their elbow joint. The normal extension of each participant was measured before throws were completed, the reason this was relevant was so that each participant's personal range of motion was taken into consideration as each person has a different level of flexibility at joints. Some values were not recorded fully as can be seen for participant F1A, so it was decided to disregard the values for their 10 yard throw to not affect the overall average of that throw for the group as a whole this is consistent for all the tables for this participant.

The extension of a normal person is around 0° but values for our participants ranged from 11° to -9°. This means for the angles that were retrieved during data analysis were very small values in comparison to other aspects like internal rotation. This led to a trend in the percentages being very large as the denominator of equation 2 was quite small. For this reason it was decided to present this section of data as the degree of extension over the maximum normal range of motion for each group of participants.

From Table 5 (appendix), Table 1 (appendix) was created so that each experience group could easily be compared, and a graph could be created for the extension of the elbow throughout each distance of throw.

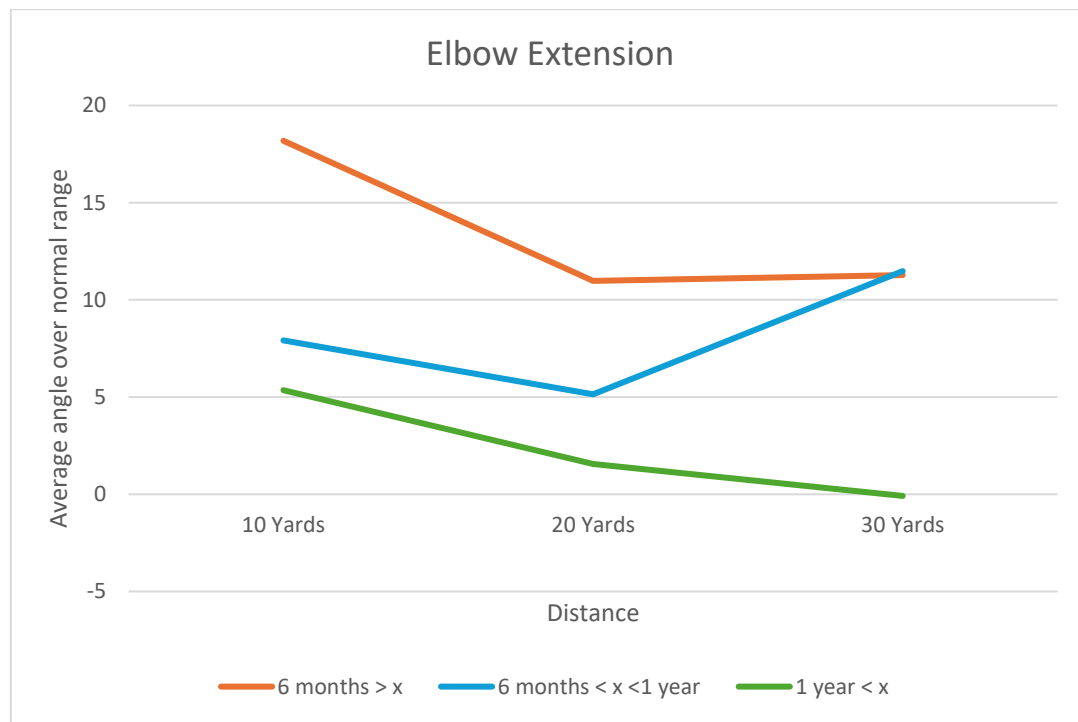


Figure 8 - Elbow extension graph.

Figure 8 represents the elbow extension data from Table 1 (appendix) so the trends can more easily be tracked. The lines on the chart represent the average values of each group across each distance. As can be taken from the graph that advanced players are the most consistent through the three throws, and on average beginners overextend the most. The commonalities between the different distances are an extension that is over three times their normal extension of their elbow. The graph also highlights that the advanced participants only overextend slightly over their normal extension. Intermediate participants have the most range in values with values ranging from 5° to 11° over their normal range across the three distances. Advanced participants extension during the throws was around their maximum extension that was calculated using the planar movements completed before the throws, their values range from -1° to 5.5° , this leads them to have the least chance of hyperextension occurring at their elbow joint during their throw. The maximum extension for each participant occurred at the end of the throw, showing hyperextension is more likely to occur at follow through stage of the throwing technique, which was expected due to results from previous research that has been studied [14].

4.2. Shoulder Flexion/Extension

Figure 9 represents the percentage that each participant over or under extends their shoulder joint. The data is presented as percentages of range of movement as seen in Table 6 (appendix), however, the normal flexion/extension of each participant was compared to the average person's range of motion found through research (180° for flexion and -60° for extension) as the planar movements of each participant was not collected. This will not affect the study too much as it can be seen from collecting values from the elbow that each participants values were very close to the values found through research.

Table 6 (appendix) and Equation 3, Table 2 (appendix) was created so that each experience group could easily be compared, and a graph could be created for the extension and flexion of the shoulder throughout each distance of throw.

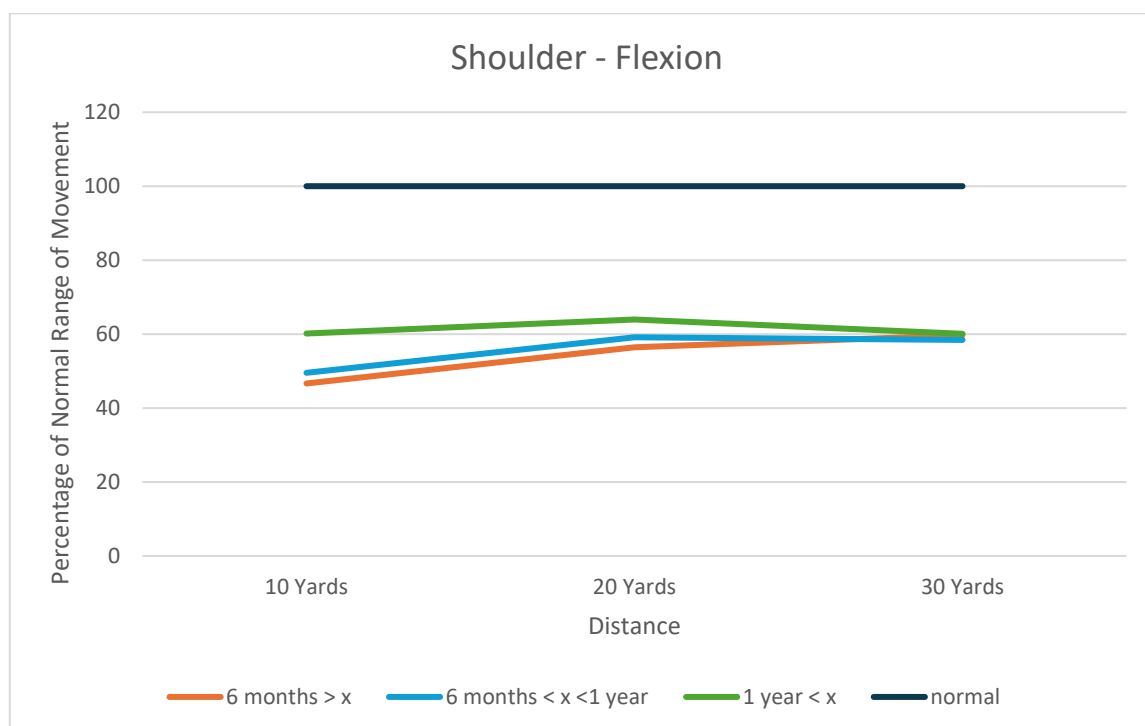


Figure 9 - Shoulder Flexion Graph

Figure 9 represents the flexion at the shoulder using data from Table 2 (appendix). The lines show the average values of each group at each distance. From the graph it highlights that each group was very consistent with the maximum flexion of the shoulder only reaching around 45%-60% of the maximum normal flexion for all experience levels. This highlights a good technique from each of the players as they should finish their throwing technique with a flexion of 90° at the shoulder.

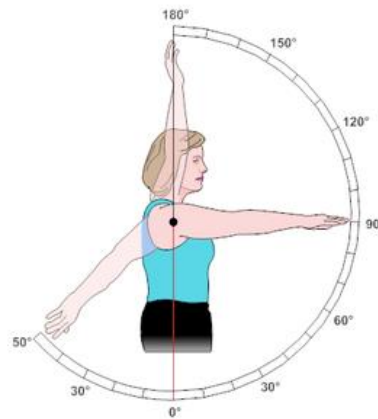


Figure 10 - shows flexion of 90° which is where a player should finish their throwing technique [27]

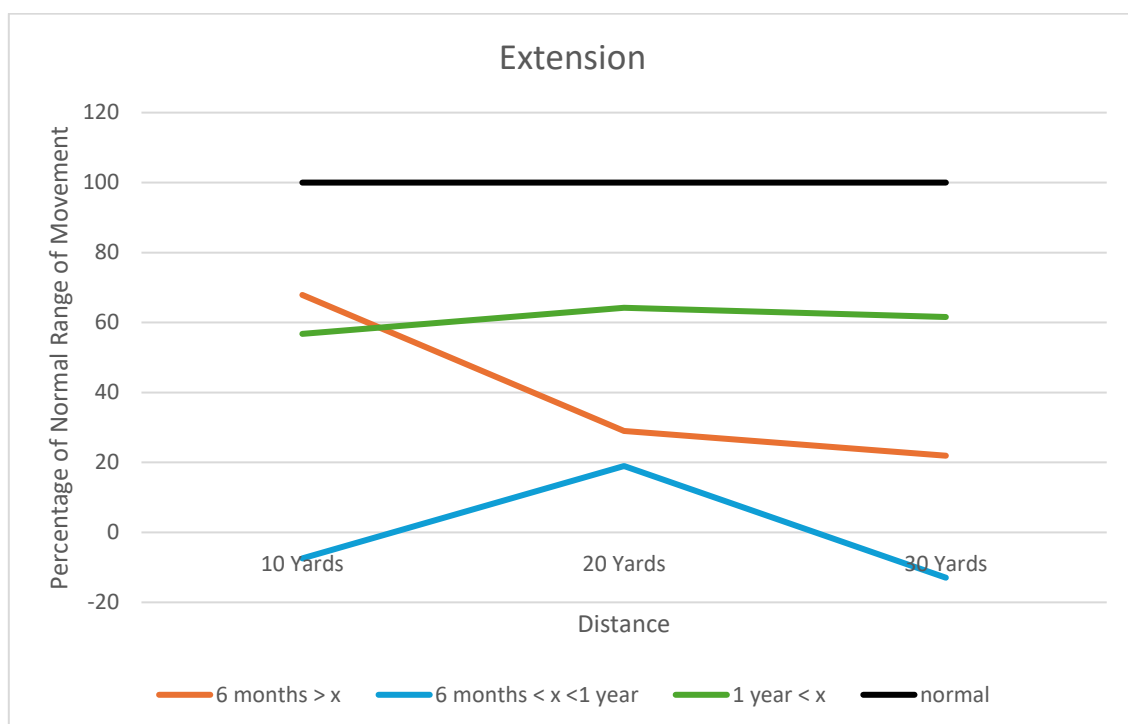


Figure 11 - Shoulder extension graph.

Figure 11 represents the extension at the shoulder using data from Table 2 (appendix). The lines on the chart represent the average values of extension of each group across each distance. The advanced group is the most consistent across the three distances with values around 60% of the normal range of motion. The other two groups were quite inconsistent with their values. However, beginners' extension decreases as the throwing distance increases, whereas the intermediate players peak at the second distance with the highest extension of 20% compared to a maximum extension of a person. Figure 9 and Figure 11 demonstrates that players are not likely to obtain an injury at their shoulder due to neither of the players extension or flexion of the shoulder reach more than a person's natural maximum extension or flexion due to good technique being taught.

4.3. Shoulder Abduction

Table 7 (appendix) represents the percentage that each participant abducts their shoulder joint. The data is presented in the same way as Table 6 (appendix), the normal abduction of each participant was compared to the average person's range of motion, which was found through research, these values 150° for abduction.

Again Table 7 (appendix) and Equation 3, Table 3 (appendix) was created so that each experience group could easily be compared, and a graph could be created for the abduction of the shoulder throughout each distance of throw.

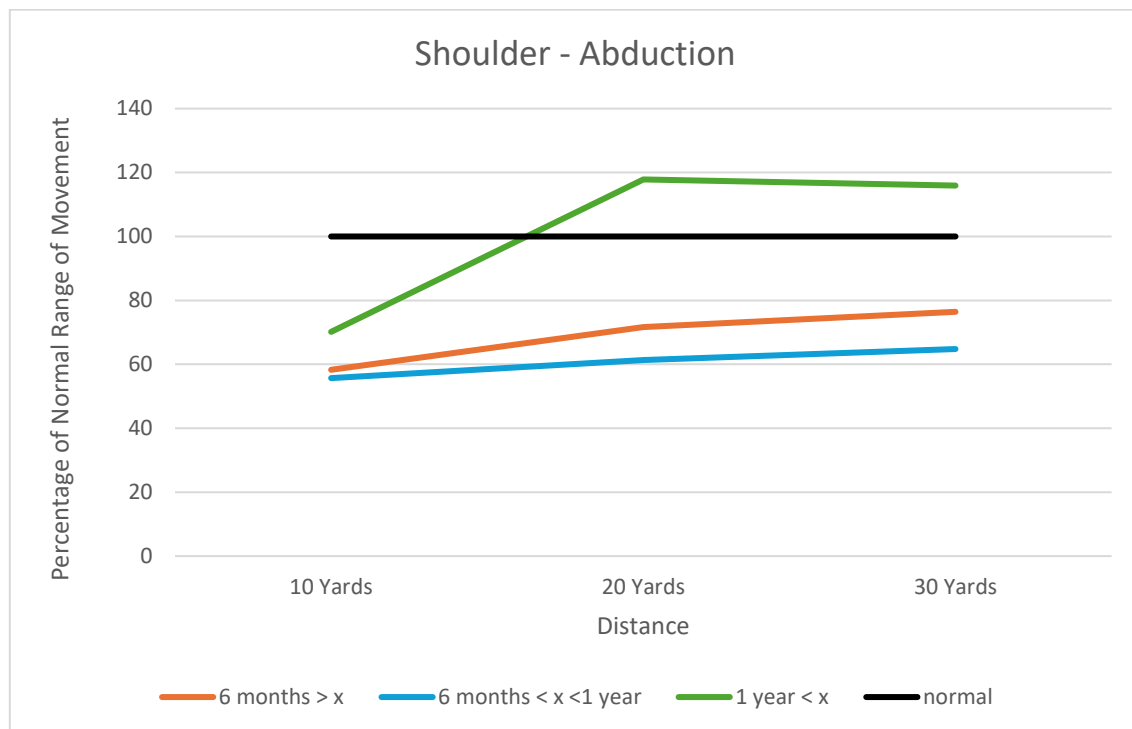


Figure 12 - Shoulder abduction graph.

Figure 12 represents abduction using data from Table 3, the lines show the average values of each group at each distance, both beginners and intermediates have consistent abduction below the maximum values for each distance, however the advanced group has an increase at 20 yards over the maximum abduction value although it stays quite consistent between 20 yards and 30 yards. The abduction is around 118% of the max abduction for the advanced player.

4.4. Shoulder External/Internal Rotation

Table 8 (appendix) represents the percentage that each participant internal and externally rotates their shoulder joint. The data is presented in the same way as Table 6 (appendix), the normal internal/external rotation of each participant was compared to the average person's range of motion, which was found through research, these values 80° for internal rotation and -90° for external rotation.

Again Table 8 (appendix) and Equation 3, Table 4 (appendix) was created so that each experience group could easily be compared, and a graph could be created for the internal rotation and external rotation of the shoulder throughout each distance of throw.

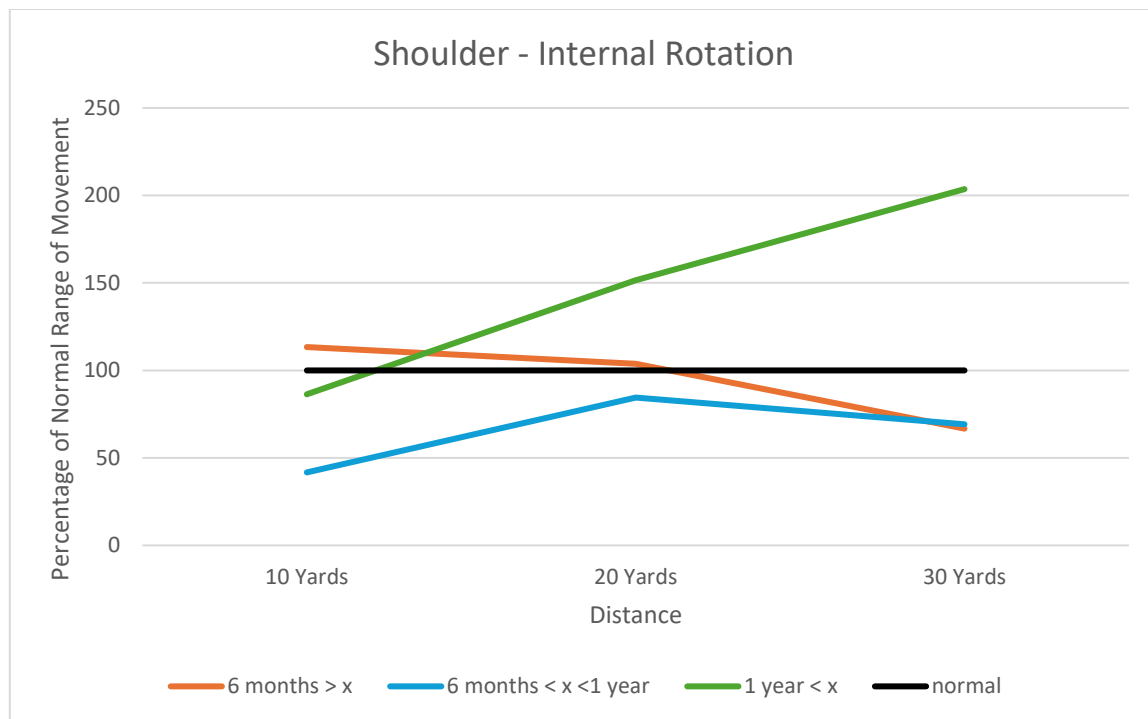


Figure 13 - Internal rotation of the shoulder graph.

Figure 13 represents internal rotation using data from Table 4 (appendix), the lines show the average values of each group at each distance, beginner participants start over rotating internally but this lowers below the maximum normal level at 30 yards, the intermediate group is quite consistent throughout each throw, always being below the maximum internal rotation. The graph's most intriguing feature is the advanced group linearly increasing above the normal through each throw increasing their risk of injury due to their technique.



Figure 14 - External rotation of the shoulder graph.

Figure 14 represents external rotation using data from Table 4 (appendix), the lines show the average values of each group at each distance, beginner participants are consistent and stay around the maximum external rotation of the shoulder slightly increasing as the distance's increases. The intermediate group is quite consistent again throughout each throw, always being below the maximum external rotation and slightly increasing as the distance gets further like the beginner group. The advanced group on is again over the maximum external rotation for each distance and over rotates more on the longer throws, the reason this over rotation is relevant is because continuous over rotation both internally and externally will cause degeneration of the tendon, leading to possible tears in the rotator cuff. It can also lead to an increased risk of dislocation of the shoulder.

4.5. Overall results

From examining the results from the experiment, it can be noted that the beginners and intermediate players had the highest risk of injury at the elbow. When inspecting the results at the shoulder joint none of the participants reached the maximum flexion/extension in the shoulder, however it was highlighted that the advanced players use a larger range of motion in the shoulder and often for larger throws exceed the normal range of motion in the shoulder. The most noticeable trends that occurred was the consistency of the technique of beginners with each distance and intermediate players being the most inconsistent with throws across the different distances.

Overall, the results appear to indicate that the beginners and intermediate groups are still learning the technique for a throw in the elbow joint, whereas their motion in their shoulder is

good. As predicted the advanced group are confident in the technique at the elbow, however an unexpected observation was that the advanced players have poor technique at the shoulder with regular over rotation and abduction of the joint. This could be due to coaching. When coaching quarterbacks the elbow joint is more exposed, therefore, it is easier for coaches to spot issues in techniques at the elbow, however from the results it appears that the technique at the shoulder then gets neglected. These findings will now be considered as a part of existing literature about the throwing techniques of quarterbacks and injuries that occur at the shoulder and elbow joints.

5. DISCUSSION

This project aimed to analyse the biomechanical movement at the shoulder and elbow during an overarm throw in American Football. By using a motion capture suit, it allowed conclusions to be if certain movements or experience have a greater the risk of injuries at these joints. It was also researched if the distance the ball was thrown had correlations with the risk of injury. The discussion section of this report is going to look at these aims and objectives, detailing the research findings alongside other literature to conclude outcomes from this study, and help coaches look at common issues in technique and how to reduce the risk of injury to their players by small changes in their technique. The main conclusions that answer the aim of this study were that most players had the basic, correct technique. However, in terms of movements the intermediate group were the least consistent when analysing the movement on the software across the different distances. It was also discovered that beginners were most likely to get injured at the elbow due to constant hyperextension of the elbow joint for each throw. In contrast, the advance players were most likely to have injury to occur at their shoulder due to over rotation of the joint.

5.1. Technique

A main component for the technique for an American football throw is the 90° angles held through the throws, these angles occur between the forearm and the humerus, at the elbow joint and also between the humerus and main body at the shoulder joint, these angles can be seen in figure 15. During a throw a player should not need flexion in their humerus to be more than 90° even though they have a range of movement of around 180° at the shoulder joint. It can be seen from Figure 9 that all participants flexion is 45%-60% of the normal range of 180°, showing on average all participants having near a 90° at their shoulder joint highlighting their good understanding of this aspect of the throw. From looking back at recordings participants in the beginner and intermediate group struggled to keep an angle of 90° at the elbow joint, as can be seen in Figure 16 for participant F2B, this means that they often lose power in their throws. Throughout the three distances there was consistency between throws for beginners and advanced players, whether it be across all three distances or across two consecutive

distances but especially for extension the intermediate group values were very varied for each distance. For example in Figure 8, the graph for extension at the elbow joint the values for the intermediate group respectively for the 10 yard, 20 yard and 30 yard throws were, 7.9° , 5.1° and 11.47° over their normal range of movement, this highlights the inconsistency of when the participants release the ball and the follow through over each distance. After speaking to intermediate players after the data collection they said for the further throws they stated that they were worried about making the distance of the throw which meant they were less concerned with completing the correct technique. To stop these inconsistencies players must focus on their technique rather than the distance they need to throw. A good training drill to get players familiar with the right technique is having a player start off with short throws, focusing on releasing the ball as it comes into the view of peripheral vision, they should focus on following the arm through until it touches the opposing hips. This drill allows players to cement how the movement should feel. As the player get more confident the distance of the throw can be increased. This exercise should be used as a warmup before training begins and matches to remind players of how the movement should feel to prevent incorrect technique during a game, as players will be more likely to feel if the technique is off.



Figure 15 - 90° angles during a throw [28]

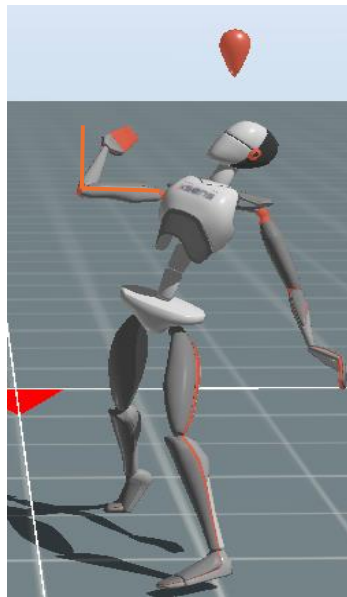


Figure 16 - F2B 30 yard movement.

5.2. Experience Level

From the study based on runners that took place over 4 years it was concluded that beginners were more likely to get injured than advanced runners [13], for this study it was expected to have the same outcome. However, this study deduced that although beginners were more likely to get injured at their elbow, advanced players were much more likely to sustain an injury at their shoulder. After speaking with coaches from the American Flag Football team at the University of Nottingham it was gathered that it was easier to spot incorrect technique in the elbow as the joint is much more exposed in comparison to the shoulder, even in video playback it is hard to see the movement in the shoulder since it is covered by more muscle and bone. Since the beginners had no issues at the shoulder it highlights the importance of going back and looking at the basics of the technique and focusing on those to prevent over rotation of the shoulder. In addition, the over rotation comes from the player trying to get more momentum in the ball, so it travels further. Players should look more into their lower body movement to increase this momentum rather than using the upper body to increase the momentum of the ball to reduce the risk of injury to the shoulder.

5.3. Distance thrown

For the study there were three distances participants were required to throw, these distances were chosen so that participants would be comfortable and also challenged. The study expected to have male participants, so it was chosen to have the final throw as 30 yards to ensure the male participants were challenged, but due to their game schedule they did not have time to partake in the study. This meant many of the females could not throw the 30 yard distance, so the final throw was a best attempt for all female participants. It was concluded

that the advance participants could throw further than all the other participants. It was concluded that the movement was different over the different distances for advanced groups and intermediate groups, this could be as they were trying to gain more momentum by increasing the rotation of the shoulders.

5.4. Injuries

In the professional American Football 14% of shoulder injuries occur due to the throwing technique of the player, [2], by looking at the flexion/extension of the elbow and shoulder and the abduction/adduction with the internal/external rotation of the shoulder the study aims to pinpoint when the movements that increase the risk of injury occurs. For the injuries that occur to a quarterback 22% of these are at the shoulder [16], examples of injuries that transpire are dislocation of the joint and tearing of ligament due to the continuous stretching of the ligament [3]. These injuries arise due to the abduction and internal/external rotation of the shoulder during a throw being greater than the normal range of motion in the shoulder. From this study, it can be highlighted that beginners and intermediate players did not exceed a person's normal range of motion for either the rotation or abduction at the shoulder. In contrast the advanced players surpassed the normal range of motion for both their 20 and 30 yard throws. Their internal rotation linearly increased across the three distances from 80% at 10 yards to 200% at 30 yards, however the external rotation decreased from 195% at 20 yards to 180% at 30 yards. In addition, the abduction graph is similar to Figure 14 for advanced players, with the abduction being around 118% for both 20 yard and 30 yard throws. The consistent over rotation while the players complete these movements in the long term will lead to the stretching of the ligaments in the shoulder, like the glenohumeral ligament that tightens to stop the over rotation of the joint, increasing the players risk of tearing the ligament or dislocation of the shoulder.

Elbow injuries to quarterbacks make up about 13% of all injuries recorded to quarterbacks [16], this is less than the amount of injuries that occur at the shoulder. Even though the elbow is a much more stable joint with only two degrees of freedom compared to the shoulders six degrees of freedom the injury rate is just as likely. The main cause of injury at the elbow is hyperextension of the joint [7]. A hyperextended joint is the excessive movement of the joint past its normal range of movement [12], damaging the tissue around the joint, which can wear tissue and cartilage away causing swelling and pain at the joint [11]. This study showed that the players most likely to sustain an elbow injury are the beginners, as they consistently overextended past their normal range of motion for all three throws. The intermediate group did also have a trend of overextending their elbow joint during the throwing motion for each throw but not to the extent that the beginners hyperextended, especially during the shortest throw which is the most performed by players. Again, to prevent these injuries players should complete the drill spoken about in the technique section of this discussion, to allow the players to warm up and their body can learn what the correct technique feels like. Muscle memory will allow for players to learn what feel correct and safe, and if a player overextends, they should

be able to feel the difference in technique, if their body knows how what the correct technique is.

5.5. Limitations

As all participants were recruited from the University of Nottingham American Football Society, only 90% of society members have not partaken in American Football before joining the university, this means that the maximum experience most players have is around three years. In many study's people with only three years of experience would not been seen as advanced players, however for this study it was decided that players with more than one years of experience would be in the advanced group, due to their only being around four quarterbacks in the society with more than a year's worth of experience. The lack of experience within the society meant that when the four members with over a years' worth of experience were approached only one chose to participate in the study. This restricted the study as anomalies in the throws completed in the advanced group were hard to highlight as there was not another participant to compare F3A to. Ideally there would have been 3 participants for each group but like in many biomechanics studies it is hard to recruit as people have busy schedules. Busy scheduling also meant that males who were interested in partaking in the study could not as the male contact teams season is busiest in February and March as they have both cup and league matches, this meant that no males could participate in the study, it is not expected that this will changes the outcomes retrieved from the study as literature studied highlighted that there were no differences in the shoulder and elbow movement between males and females [10]. However, male participants would have been able to throw the 30 yard distance unlike the female participants so it would have been informing to have had them participate.

For the longest throw, that was the best effort for female participants, the distances reached by female participants were 20-26 yards, the beginner group especially struggled with the longer throw only reaching the 20 yard line on the pitch, this meant that their technique was the same the 20 yard throw and best effort throw which could explain the consistency between the two final throws for beginners on the graph. Preferably, the study would have included more throws with smaller increments, for example, 5 yards, 10 yards, 15 yards, 20 yards and 25 yards, this would have allowed for results to be more clear, and better see at what point players techniques begin to change slightly, but would also provide a challenge for the more experienced players. In addition, the planar movements of each participant were not recorded for the shoulder, this meant unlike the elbow where each participant was compared to their own personal range of motion, the percentage of range of motion was calculated using the averages found from other literature, [18]. This limits the study, as we found from measuring each participant's range of motion in the elbow there was a range from 9.3° to -11° for extension. This is $\pm 11^{\circ}$ from the average humans, it should not impact the outcome of the study all of the over rotation, that causes injury at the shoulder was more than $\pm 10^{\circ}$, but it would have made the study more accurate to have recorded the normal range of motion for each participant at the shoulder as well as the elbow.

The study originally aimed to take data from a total of ten throws and averages of throws three to seven would be taken as this would mean the participant has warmed up into the throws, but the final few throws could be affected by fatigue. All participants did complete these throws; however, the data was lost after the second throw for the majority of participants. Movella was contacted but they explained there was no way to recover the lost data. This should not affect the study outcomes and the aim of the study still has been achieved but more throws would have allowed the study to spot anomalies and disregard them.

6. CONCLUSIONS

Based on evidence from this study, it can be concluded that beginners and advanced players are most likely to be at risk of injury through an overarm American football throw. It can be deduced that beginners are most at risk of injury at the elbow due to continuous over extension of the joint no matter the length of throw. By implementing the suggestion for completing a basic throwing drill before training and games will help reinforce the correct technique for beginners. Additionally, even though advanced players are trained out of hyperextending their elbow joint due to the exposed nature of the joint and accessibility of coaches correct issues at this joint, players pick up bad habits at the shoulder joint. Often over rotating the joint for further throws is due to the player trying to increase the momentum in the ball due to the increased power needed to travel a further distance. Instead, advanced players should focus on their basic technique and stopping their throw after releasing the ball. In conclusion, it is important that all players focus on the basics of a throw, and ensuring athletes implement the suggestions from this study will lower the risk of injury to players and make players more comfortable with the movements their body can safely do.

7. FUTURE WORK

With the findings of this study indicating a greater risk of injury for throws of greater distances, an avenue for further research is finding out at what distance the risk becomes too high. Future studies could have more distances with smaller increments than those used in this study, to highlight at what point the range of motion of the throw is greater than the normal range of motion. Additionally, future studies could look to recruit players from teams outside of university so that they have a much greater experience level and also increase the sample size. This study focused on the upper body movement but as we learnt players that used footwork had much more power in their throws so it would be beneficial for the American football community to look how the body works as a whole to optimise the throwing technique. It would also be beneficial to look at how fatigue affects players throws throughout a game. The goal of any future study would be to further understand the biomechanical movement during a throw and how injuries can occur due to some of these movements.

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9. Appendix

9.1. Results Tables

Table 1 - Elbow extension averages for each group

	10 Yards (average angle over max range in °)	20 Yards (average angle over max range in °)	30 Yards (average angle over max range in °)
Less than 6 months - Beginner			
6 months > x	18.2	11.0	11.3
More than 6 months, less than a year - Intermediate			
6 months < x < 1 year	7.9	5.1	11.5
More than a year - Advanced			
1 year < x	5.4	1.6	-0.1

Table 2 - Shoulder flexion and extension averages of each group

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Less than 6 months – Beginner			
Flexion	46.7	56.5	59.7
Extension	67.9	29.0	21.9
More than 6 months, less than a year – Intermediate			
Flexion	49.5	59.1	58.4
Extension	-7.4	19.0	-13.0
More than a year – Advanced			
Flexion	60.1	63.9	60.1
Extension	56.8	64.2	61.6

Table 3 - Shoulder abduction and adduction average of each group

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Less than 6 months – Beginner			
Abduction	58.3	71.6	76.4
More than 6 months, less than a year – Intermediate			
Abduction	55.7	61.4	64.8
More than a year – Advanced			
Abduction	70.2	117.8	115.9

Table 4 - External and internal rotation averages of each group of the shoulder

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Less than 6 months – Beginner			
Internal Rotation	113.5	103.7	66.8
External Rotation	84.9	98.0	100.3
More than 6 months, less than a year – Intermediate			
Internal Rotation	41.7	84.5	69.3
External Rotation	68.4	73.0	84.9
More than a year – Advanced			
Internal Rotation	86.3	151.5	203.6

External Rotation	117.2	195.8	181.2
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Table 5 - Elbow extension of each participant

	10 Yards (angle over max range in °)	20 Yards (angle over max range in °)	30 Yards (angle over max range in °)
Participant F1A			
Extension	N/A	17.8	10.2
Participant F1B			
Extension	18.2	4.1	12.4
Participant F2A			
Extension	12.2	14.5	22.5
Participant F2B			
Extension	10.7	2.5	17.8
Participant F2C			
Extension	0.9	-1.6	-5.8
Participant F3A			
Extension	5.4	1.2	-0.1

Table 6 - Shoulder flexion and extension of each participant

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Participant F1A			
Flexion	N/A	68.9	68.9
Extension	N/A	-42.8	6.5
Participant F1B			
Flexion	46.7	44.0	50.5
Extension	67.9	100.8	37.3
Participant F2A			
Flexion	55.1	60.7	85.4
Extension	-95.7	-71.9	-88.3
Participant F2B			
Flexion	57.0	81.0	58.7
Extension	54.3	117.8	8.6
Participant F2C			

Flexion	36.5	35.7	31.2
Extension	19.1	11.0	40.9
Participant F3A			
Flexion	60.1	63.9	60.1
Extension	56.8	64.2	61.6

Table 7 - Shoulder abduction of each participant

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Participant F1A			
Abduction	N/A	83.4	93.5
Participant F1B			
Abduction	58.3	59.9	59.4
Participant F2A			
Abduction	64.04	69.9	84.9
Participant F2B			
Abduction	58.55	67.1	64.5
Participant F2C			
Abduction	44.5	47.1	45.0
Participant F3A			
Abduction	70.2	117.8	116.0

Table 8 - External and internal rotation of the shoulder of each participant

	10 Yards (% of normal range)	20 Yards (% of normal range)	30 Yards (% of normal range)
Participant F1A			
Internal Rotation	N/A	48.7	43.0
External Rotation	N/A	103.6	103.3
Participant F1B			
Internal Rotation	113.4	158.8	90.5
External Rotation	84.9	92.4	97.3
Participant F2A			
Internal Rotation	34.3	83.1	43.3
External Rotation	74.7	71.3	108.1
Participant F2B			
Internal Rotation	78.1	122.1	97.4
External Rotation	86.7	86.7	95.9
Participant F2C			
Internal Rotation	12.8	48.2	67.1

External Rotation	44.0	61.1	50.6
Participant F3A			
Internal Rotation	86.4	151.5	203.6
External Rotation	117.2	195.8	181.2

9.2. Ethical Forms

Consent Form

Title of Study: Understanding the Biomechanics of the upper body during an over arm American football throw.

Name of Researcher(s): [REDACTED]

This form is to help you to decide whether or not you would like to take part in this study. Please sign at the bottom should you feel that you agree with all the following statements.

	Yes	No
I have had the opportunity to consider the participant information sheet, ask questions and have had these answered satisfactorily.		
I understand that I can refuse to take part if I wish, without giving a reason and my refusal will not affect my ability to take part in future studies.		
I understand that I can withdraw from the study at any time without giving a reason without prejudice. In doing so, I know that any data collected by this point will be permanently erased.		
I know that I can ask the researcher for further information about the study at any time.		
I understand that all information and data I give will be confidential and anonymised , and that it will not be possible to identify any of the respondents in the study report.		
I understand the risks associated with the physical activities involved in this study.		
I consent to the information and data collected from my participation to be used for this purpose of completing a final year dissertation project		
I consent to the data being stored securely and used for the current and future research studies with results potentially being published in academic journals and presented at scientific research conferences.		
I agree to take part in the above study.		

Participant signature:	Date:
Researcher signature: [REDACTED]	Date: 6/12/23

Participant Information Sheet

Title of Study: Understanding the Biomechanics of the upper body during an over arm American football throw.

Name of Researcher(s): [REDACTED]

I would like to invite you to take part in my research study. Before you decide we would like you to understand why the research is being done and what it would involve for you. Please have a read of the following and do not hesitate to ask me if there is anything that is not clear.

What is the purpose of the study?

The study is for my BEng individual project. The results aim to help to look at the biomechanics of the upper body specifically the elbow during an American football throw and specify whether certain movements could result in a higher risk of injuries.

Why have I been invited?

You are being invited to take part because you are a part of the University of Nottingham American Football Society or play a sport at the university.

Do I have to take part?

It is up to you to decide whether or not to take part. If you do decide to take part, you will be given this information sheet to keep and be asked to sign a consent form. If you decide to take part, you are still free to withdraw at any time and without giving a reason. This would not affect your legal rights.

What will I have to do?

I would ask you to come to the 3G pitch at the David Ross sports village at the University Park campus for 30 minutes –1-hour to carry out 10 overarm American football throws at three different distances (10 yards, 20 yards and 30 yards). In separate tests, I will also ask you to flex your elbow as close to your body as possible, then extend your elbow down as far as you can.

Full address – David Ross Sports Village, Beeston Ln, Lenton Abbey, Nottingham, NG7 2RD

You will be required to wear a motion capture suit for the duration of the data collection session. The session may also be filmed for post-analysis clarification of movement. Once we establish that the motion capture suit is working, you will be asked to carry out 10 throws at three distances of 10 yards, 20 yards and 30 yards. The movement will be demonstrated to you at the beginning of the session. Once you have successfully completed the 30 throws and had your maximum extension and flexion of the elbow measured, you will be free to remove the suit.



What data is taken from me?

Name, sex assigned at birth, height and weight will be collected before completing the movements. Motion capture recording your movement will be recorded to collect data including joint angles, position and stance. Video recordings may also be used for post-analysis clarification of movement. Your email and phone number may be kept for contacting you to organise the data collection session.

Before coming in:

You do not need to prepare anything prior to attending the session. Please prepare as you would for a normal training session. Should you feel you are more comfortable having someone come in with you while you participate, please feel free to have them accompany you.

What to wear/bring:

Please wear comfortable, sporty clothes, with any trainers as you would usually wear to take part in physical activity, training, or a sporting fixture. If possible, please wear tight fitting clothing. Water bottle is advised also.

What are the possible benefits of taking part?

Once the data is analysed, each participant will be offered the option to have their data sent to them. Your participation will also help increase our knowledge around American Football and possible areas in performance optimisation or injury prevention, and hopefully increase awareness around the sport university wide.

What if there is a problem?

If you have a concern about any aspect of this study, you should ask to speak to a member of the research team who will do their best to answer your questions. If you remain unhappy and wish to complain formally, you can do this by contacting the Faculty of Engineering Research Committee. All contact details are given at the end of this information sheet.

Will my taking part in the study be kept confidential?

We will follow ethical and legal practice and all information about you will be handled in confidence. If you join the study, the data collected for the study will be looked at by authorised persons from the University of Nottingham who are organising the research. They may also be looked at by authorised people to check that the study is being carried out correctly. All will have a duty of confidentiality to you as a research participant and we will do our best to meet this duty. All information which is collected about you during the course of the research will be kept strictly confidential, secured within the University of Nottingham. Any information about you which leaves the University will have your name and address removed (anonymised) and a unique code will be used so that you cannot be recognised from it. Anonymised data may also be stored in data archives for future researchers interested in this area. Your personal data will be kept for 3 months after the end of the study so that we are able to contact you about the findings of the study and possible follow-up studies (unless you advise us that you do not wish to be contacted). All identifiable research data will be kept securely for 7 years. After this time your data will be disposed of securely. During this time all precautions will be taken by all those involved to maintain your 2023-24 academic year only confidentiality, only members of the research team will have access to your personal data.

What will happen if I don't want to carry on with the study?

Your participation is voluntary, and you are free to withdraw at any time, without giving any reason, and without your legal rights being affected. If you withdraw, you may request the data to be erased.

What will happen to the results of the research study?

The research will be completed by the end of May 2024, at which time, personal data can be requested and/or a copy of the paper sent to you. The paper will be uploaded online and will be examined as part of my BEng final grade.

Who has reviewed the study?

This study has received a favourable ethical opinion by the Faculty of Engineering Ethics Committee.

Further information and contact details

Project Researcher: [REDACTED]

Project Supervisor: Dr Samuel Smith, Assistant Professor, Department of Mechanical, Materials and Manufacturing Engineering, Faculty of Engineering ezzshs@exmail.nottingham.ac.uk

Senior Research Administrator: [REDACTED]